

# Trading Bot — System Architecture

Wyckoff Range Strategy · Paper Trading POC · v5

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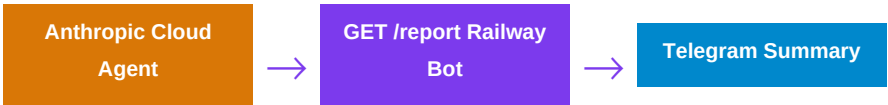
## 1. System Overview

A fully cloud-hosted automated paper trading system. TradingView Pine Script detects Wyckoff range setups on 15-minute charts and fires webhook alerts to a Flask bot running on Railway. The bot executes orders on an OANDA demo account (forex pairs), logs all data to a Railway PostgreSQL database, and notifies via Telegram. A scheduled Anthropic cloud agent monitors the bot daily. **No local machine is required for any part of the system to operate.**

### Live Signal Flow



### Daily Monitoring Flow (7am CT)



### Deploy Flow



## 2. Where Everything Lives

| Component           | Platform | URL / Location                                | Always On? | Local Machine? |
|---------------------|----------|-----------------------------------------------|------------|----------------|
| Bot Runtime (Flask) | Railway  | tradingbot-production-1e5a.up.railway.ap<br>p | ■ 24/7     | ■ No           |
| Source Code         | GitHub   | github.com/wasiheider/trading_bot             | ■          | ■ No           |

|                     |                                    |                                                      |             |      |
|---------------------|------------------------------------|------------------------------------------------------|-------------|------|
| Database            | <a href="#">Railway PostgreSQL</a> | Same Railway project — DATABASE_URL injected         | ■ 24/7      | ■ No |
| Env Vars / Secrets  | <a href="#">Railway Variables</a>  | trading_bot service → Variables tab                  | ■           | ■ No |
| TradingView Alerts  | <a href="#">TradingView Cloud</a>  | 12 × 15M charts, paper-trading alerts                | ■ 24/7      | ■ No |
| TV Charts Open 24/7 | <b>Vultr VPS</b>                   | 207.148.7.167 · Dallas · \$5/mo                      | ■ 24/7      | ■ No |
| Order Execution     | <a href="#">OANDA Demo API</a>     | api-fxpractice.oanda.com · Acct 101-001-39435783-001 | ■ 24/7      | ■ No |
| Notifications       | <a href="#">Telegram</a>           | Bot token + chat ID in Railway env vars              | ■           | ■ No |
| Daily Monitor Agent | <a href="#">Anthropic Cloud</a>    | claude.ai/code/routines · 7am CT daily               | ■ Scheduled | ■ No |
| CLAUDE.md / README  | <b>GitHub</b>                      | In repo root — loaded automatically by Claude Code   | ■           | ■ No |

★ Zero local dependencies — delete the local folder and the system keeps running.

### 3. Railway Project Layout (heartfelt-serenity / production)

| Service     | Status   | Volume                     | Key Detail                                  |
|-------------|----------|----------------------------|---------------------------------------------|
| trading_bot | ■ Online | trading_bot-volume → /data | Flask app · paper-trading branch · US West  |
| Postgres    | ■ Online | postgres-volume            | DATABASE_URL auto-injected into trading_bot |

### 4. PostgreSQL Database Schema

| Table     | Primary Purpose                                          | Key Columns                                                      |
|-----------|----------------------------------------------------------|------------------------------------------------------------------|
| trades    | Full trade ledger — every entry and outcome              | instrument, direction, setup, result, pnl, oanda_trade_id        |
| bot_state | Daily/weekly PNL, balance, win/loss counters (1 row)     | account_balance, daily_pnl, weekly_pnl, total_wins, total_losses |
| signals   | Every incoming TradingView webhook — full pipeline trace | symbol, direction, setup, entry_price, sl, tp1, tp2, rr_to_tp1   |

All DB operations route through db.py — never write directly to tables from other modules.

## 5. Strategy — v5 (Active)

| Model                        | Setup Field           | Trigger                                                       | Execution                         |
|------------------------------|-----------------------|---------------------------------------------------------------|-----------------------------------|
| <b>A — BOS + Rejection</b>   | "bos" or ""           | 15M BOS near boundary → retest → rejection candle             | OANDA (forex) / Telegram (others) |
| <b>B — Spring / Upthrust</b> | "spring" / "upthrust" | Wick beyond boundary closes back inside; deferred entry       | OANDA (forex) / Telegram (others) |
| <b>C — BOS + RSI Div</b>     | "bos_div"             | Same as A + RSI divergence at retest (30–70); priority over A | OANDA (forex) / Telegram (others) |
| <b>D — Mid BOS + Retest</b>  | "mid_bos"             | 15M structural BOS within $\pm 35\%$ of range mid → retest    | OANDA (forex) / Telegram (others) |

| Parameter         | Value                                                                |
|-------------------|----------------------------------------------------------------------|
| Range logic       | 4H Pivot-Anchored — levels lock until broken by 30% of range size    |
| Quality Gate      | Boundary touches $\geq 3$ · Mid crosses $\geq 3$ · 30M swing pattern |
| TP1               | 1:1 RR — close 50% + SL moves to breakeven                           |
| TP2               | 1:3 RR — close remaining 50%                                         |
| Trailing SL       | After TP1: at 1.5R peak → move to 1R, then trail 0.5R behind peak    |
| Risk per trade    | \$250 (0.25% of \$100k account)                                      |
| Daily loss limit  | \$4,000 (4%)                                                         |
| Weekly loss limit | \$10,000 (10%)                                                       |
| Pine Script       | pine_script_paper_v5.pine — 15M chart, Once Per Bar Close alert      |

## 6. Instruments

| Instrument                                        | OANDA Execution | Telegram + Log | Note                                             |
|---------------------------------------------------|-----------------|----------------|--------------------------------------------------|
| EURUSD, GBPUSD, USDJPY, EURNZD, NZDUSD            | ■ Yes           | ■ Yes          | Forex pairs — full execution                     |
| XAUUSD, XAGUSD, US100, US30, US500, USOIL, BTCUSD | ■ No            | ■ Yes          | CFDs/metals/indices — OANDA demo doesn't support |

## 7. API Endpoints

| Endpoint       | Method | Description                                                    |
|----------------|--------|----------------------------------------------------------------|
| /              | GET    | Health check — returns strategy version + CT time              |
| /dashboard     | GET    | Live trading dashboard (HTML)                                  |
| /state         | GET    | Balance, PNL, open trades, per-model stats (A/B/C/D)           |
| /trades        | GET    | Full trade ledger from PostgreSQL                              |
| /report        | GET    | Daily monitor — sends Telegram summary (called by cloud agent) |
| /news          | GET    | Yahoo Finance RSS headlines (15-min cache)                     |
| /calendar      | GET    | ForexFactory economic calendar (1-hour cache)                  |
| /webhook/paper | POST   | TradingView signal receiver — entries + lifecycle events       |
| /admin/reset   | POST   | Full state reset — requires PAPER_WEBHOOK_TOKEN                |

## 8. Source File Map ([github.com/wasiheider/trading\\_bot](https://github.com/wasiheider/trading_bot))

| File                             | Role                                                                               |
|----------------------------------|------------------------------------------------------------------------------------|
| <b>main.py</b>                   | Entry point — scheduler (midnight reset, weekly summary, heartbeat) + Flask server |
| <b>server.py</b>                 | Flask app — all endpoints, webhook handler, /report monitor                        |
| <b>db.py</b>                     | PostgreSQL layer — all CRUD for trades, bot_state, signals tables                  |
| <b>risk.py</b>                   | Position sizing, PNL limit checks, in-memory paper_state backed by PostgreSQL      |
| <b>oanda.py</b>                  | OANDA REST API — place/close orders, open trades, account summary                  |
| <b>config.py</b>                 | All environment variables and risk constants                                       |
| <b>notifier.py</b>               | Telegram send helper — parse_mode: HTML                                            |
| <b>logger.py</b>                 | Legacy SQLite schema — not actively wired in current flow                          |
| <b>pine_script_paper_v5.pine</b> | Active Pine Script — load on all 12 × 15M TradingView charts                       |
| <b>dashboard.html</b>            | Single-file live dashboard served at /dashboard                                    |
| <b>CLAUDE.md</b>                 | Auto-loaded by Claude Code — full project brief and constraints                    |
| <b>README.md</b>                 | Public project documentation                                                       |

## 9. Environment Variables (stored in Railway — never in code)

| Variable            | Description                                                        |
|---------------------|--------------------------------------------------------------------|
| DATABASE_URL        | PostgreSQL connection string — auto-injected from Postgres service |
| OANDA_API_TOKEN     | OANDA demo API key                                                 |
| OANDA_ACCOUNT_ID    | 101-001-39435783-001                                               |
| OANDA_BASE_URL      | https://api-fxpractice.oanda.com                                   |
| TELEGRAM_BOT_TOKEN  | Telegram bot token                                                 |
| TELEGRAM_CHAT_ID    | Telegram chat ID for notifications                                 |
| PAPER_WEBHOOK_TOKEN | Shared secret — must match Pine Script token input                 |
| PAPER_ACCOUNT_SIZE  | 100000 (starting demo balance)                                     |
| DATA_DIR            | /data — Railway Volume mount path (legacy JSON fallback)           |
| ANTHROPIC_API_KEY   | Reserved for future Claude integration                             |

## 10. Automation Summary — What Runs Without Human Intervention

| Function               | Trigger                        | Automated?        |
|------------------------|--------------------------------|-------------------|
| Signal detection       | TradingView 15M bar close      | ■ Fully automated |
| Webhook delivery       | TradingView alert fires        | ■ Fully automated |
| Risk check + sizing    | Every incoming signal          | ■ Fully automated |
| OANDA order placement  | Signal approved                | ■ Fully automated |
| TP1/TP2/SL management  | TradingView lifecycle alert    | ■ Fully automated |
| Telegram notifications | Every trade event              | ■ Fully automated |
| Daily PNL reset        | Midnight CT (scheduler)        | ■ Fully automated |
| Weekly PNL reset       | Monday midnight CT             | ■ Fully automated |
| Weekly summary         | Friday 3:50pm CT (scheduler)   | ■ Fully automated |
| Daily monitor report   | 7am CT (Anthropic cloud agent) | ■ Fully automated |
| Database persistence   | Every trade / state change     | ■ Fully automated |
| Code deployment        | git push to paper-trading      | ■ Fully automated |

|                         |                                |                      |
|-------------------------|--------------------------------|----------------------|
| Prop firm trade entry   | Manual — FTMO/Apex have no API | ■■ Manual — always   |
| TradingView alert setup | When Pine Script is updated    | ■■ Manual — one-time |